Cold-Weather Burst Rates in Iron Mains

H. Fixture J. Fixture

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FICTIONAL TEST MATERIAL: this paper does not exist; it was written for the citation-audit fixture. No author, utility, venue, or measurement described here is real.

Abstract

Iron water mains break more often in winter. We combined eight years of burst records from a fictional utility with soil-temperature measurements at the depth of the mains. Burst rates in cast-iron and ductile-iron mains were 2.4 times higher on days when the soil temperature was below 2 °C than on days when it was above 2 °C. We describe the threshold, its stability across pipe classes, and its use for planning winter repair crews.

1 Introduction

Cold weather stresses buried iron mains in two ways. Cold water contracts the pipe and raises the axial stress at joints and corrosion pits, and freezing soil shifts and loads the pipe. Utilities know that bursts cluster in winter, but they lack a simple, measurable trigger for when the risk rises.

This paper relates burst rates to the soil temperature at the depth of the mains, rather than to the air temperature, and identifies a threshold that separates low-risk from high-risk days.

2 Method

2.1 Burst records

We used eight years of repair records from the fictional Fixture Heights utility. A burst is a main break that required an unplanned repair. Only cast-iron and ductile-iron mains were included. Burst counts were normalised by the length of main in service on each day.

2.2 Soil temperature

Six soil-temperature stations recorded the temperature at 1.2 m, the typical depth of the mains, every hour. Each burst was matched to the daily mean soil temperature at the nearest station.

2.3 Threshold analysis

We compared burst rates above and below candidate thresholds from −2 °C to 6 °C in steps of 0.5 °C and chose the threshold that best separated the two groups.

3 Results

3.1 Burst-rate ratio at 2 °C

The best threshold was a soil temperature of 2 °C. Burst rates in cast- and ductile-iron mains were 2.4 times higher when the soil temperature was below 2 °C than when it was above it. In other words, burst rates in these iron mains more than doubled on the cold side of the threshold.

3.2 Stability across pipe classes

The ratio was similar for cast iron and ductile iron, and for small and large diameters. It held in each of the eight winters of the record, although the number of days below the threshold varied widely from winter to winter.

3.3 Planning use

Because soil temperature changes slowly, a forecast of the days below 2 °C gives the utility several days of warning to schedule additional repair crews.

4 Limitations

This study analyses burst rates against soil temperature only.

This study does not examine chlorine or any other aspect of water quality: no water samples were analysed. This study does not examine leak localization: we used the repair records as given and did not consider how the bursts were found or located. The analysis covers iron mains only; plastic mains respond to cold differently and were excluded. The soil stations were few, so the temperature matched to a burst may differ from the temperature at the exact burst site.